

CNS neuroglia cells : Don't transmit signals

① Astrocytes

- help in formation of Blood Brain Barriers
- important for brain scar tissue formation
- share in metabolism of neurotransmitters تشارك في التمثيل الغذائي للنواقل العصبية
- convert the glucose to lactate for neuronal nourishment

② Oligodendrocytes

- formation of myelin sheath around CNS neurons

③ microglia

- phagocyte cells

④ Ependymal cells

- produce cerebrospinal fluid

* the cells in sense organs (أعضاء الحس المسؤولة عن الحس حواس)

is → bipolar (have one axon & one dendrite) عشان تكون سوية

Sensory Receptors

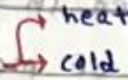
stimulation of sensory receptors by

- ① Mechanical stimuli ② chemicals ③ temperature ④ Light

The sensory Receptors in the body classified into

- ① Mechano receptors
- skin tactile receptor مستقبلات اللمس الجلدية
 - Deep tissue sensibility حساسية الأنسجة العميقة
 - Hearing receptor مستقبلات السمع
 - Baroreceptors (pressoreceptors) in Heart
 ↳ regulate the arterial blood pressure

② Thermo receptor

- sensitive to temperature changes 

③ Noci receptor

- sensitive to painful stimuli

④ Electromagnetic (photo) receptor

- sensitive to light, found in retina of eye
- Responsible for vision

⑤ chemo receptors

- sensitive to O_2 & CO_2 tension, glucose, amino acids and fatty acids
- found in taste buds & nasal mucosa
- ⊗ glucose receptors → found in the hunger & satiety centers in hypothalamus

Pain → is a protective mechanism for the body

Types of pain

الألم الحاد

① Acute pain

- occurs in 0.1 second & could be sharp pain, electrical pain
- conducted through A(α) fibers
- velocity of 6 - 30 meters/second
- terminate at reticular formation of medulla, pons & may be reached to thalamus, hypothalamus, cerebrum

الألم البطيء

② slow pain

- occurs in several second or minutes & called (burning pain)
- conducted through E fibers
- velocity of 0.5 - 2 meters/second
- terminate at reticular formation of brain stem

- Control of pain (Analgesia system) (Brain opiate system)
- chemical substances secreted in the brain and reduce the feeling of pain (similar to action of morphine on nerves) called → morphine like substances

↓ Endorphins ↓ Enkephalins ↓ Dynorphins

Thermal sensation

Sensation of cold or Heat depends on → presence of cold Receptors or warm receptors on the skin

Degree of sensation depends on →

- power of heat (warm, Hot, burning)
- power of cold (cool, cold, Freezing)

Spinal cord

- slender cylinder as finger composed of → central gray area
- peripheral white area

Gray matter

- Filled with cell bodies & dendrites
- cell bodies clustered together and form nuclei (centers)
- lack of myelin sheath

white matter

- made of nerve axons (transmit action potentials inside spinal cord or brain)
- nerve axon form bundles called (tracts) → transmit signals from spinal cord → brain
- afferent fibers enter the spinal cord called (dorsal roots)
- efferent fibers leave the spinal cord called (ventral roots)
- dorsal & ventral roots combine together to form → spinal nerve
- have Myelin Sheath

Spinal cord reflexes

① stretch reflex

- muscle spindles → transmit information about muscle length
- tendons → responsible for tension of muscle

② Knee jerk reflex

نفرب وتر الرفقة
بشاكوش

striking of patellar tendon with a hammer

③ Flexor reflex (nociceptor reflex) (pain reflex)

stand on one leg and then switch to another leg (cross extensor reflex)

Brain

consist of → ① Brain stem ② cerebrum ③ cerebellum

Brain stem : composed of → medulla, pons and midbrain

↳ function: link between spinal cord & brain

- origin of Vagus nerve (largest parasympathetic nerve)
- many branched neurons clustered together to make

البلع 1 swallowing 2 cardiovascular 3 respiratory

القيء 4 vomiting 5 cough centers
محة

• Cerebrum: the highest place in brain

- composed of → cortex, basal ganglia, Thalamus and hypothalamus

- have cerebral blood flow (responsible for consciousness) مسؤول عن الوعي

↳ if decreased → it will cause dizziness لو نقصن هيسبب دوخة في الرأس

Cerebral Blood Flow

حاجات تؤثر على

1- glucose level

1- Acidosis & Alkalosis (PH)

2- CO₂

3- O₂

4- Vasoconstriction & Vasodilation